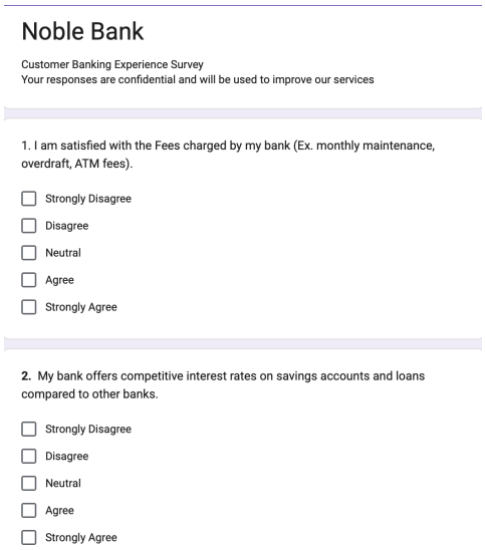
DA 233 Applied Information Management Systems
Assignment II

Part 1 ERD

For the survey, please have at least 8 items, excluding account IDs and demographic data such as customer names, account IDs, addresses, telephone numbers, email addresses, etc. You can be creative about the items, which will become fields in a database. Please make sure the data can be incorporated into a database and are useful for analysis.
The survey must be neatly laid out but there is no need to focus unnecessarily on the design aspects since we did not cover it in class. The objective is to have relevant survey items that can be incorporated in a database and be useful for analysis. Include a short explanation of (1) why you chose these survey items and (2) how they will be measured (if not immediately apparent and/or if applicable). The rationale for the survey items must be based on the problem given below. There is no need to actually collect the data.
For the ERD, please include at least 4 relevant entities generated from your survey items. For instance, two survey items, such as customer name and telephone number can become a single entity for customer. You may include customer as an entity. There is no need to have all possible attributes. But please have relevant/suitable ones for each entity. Some attributes can be pointless, depending on the design of your database. And we don't want to have unnecessary ones. Generally, each entity should have at least 2 attributes. You may choose not to have a primary key for any particular attribute if it doesn't apply. Think about the problem below and what data (and hence information) are relevant. You are to use Visio (or equivalent) and paste the ERD in a word document, along with the survey. You can be creative with the entities.
For the survey and ERD, there is no perfect answer but there are clearly solutions that are better than others.

For the survey, I did it on **Google Forms** which includes a total of ten questions.



Noble Bank

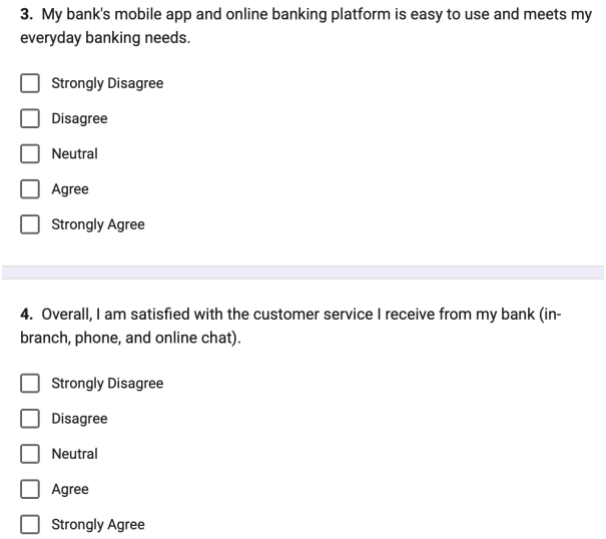
Customer Banking Experience Survey
Your responses are confidential and will be used to improve our services

1. I am satisfied with the Fees charged by my bank (Ex. monthly maintenance, overdraft, ATM fees).

- [ ] Strongly Disagree
- [ ] Disagree
- [ ] Neutral
- [ ] Agree
- [ ] Strongly Agree

2. My bank offers competitive interest rates on savings accounts and loans compared to other banks.

- [ ] Strongly Disagree
- [ ] Disagree
- [ ] Neutral
- [ ] Agree
- [ ] Strongly Agree



3. My bank's mobile app and online banking platform is easy to use and meets my everyday banking needs.

- [ ] Strongly Disagree
- [ ] Disagree
- [ ] Neutral
- [ ] Agree
- [ ] Strongly Agree

4. Overall, I am satisfied with the customer service I receive from my bank (in-branch, phone, and online chat).

- [ ] Strongly Disagree
- [ ] Disagree
- [ ] Neutral
- [ ] Agree
- [ ] Strongly Agree

5. How likely are you to recommend your current bank to a friend or family member? (1 = Very Unlikely → 5 = Very Likely)

- [ ] 1 = Very Unlikely
- [ ] 2 = Unlikely
- [ ] 3 = Neutral
- [ ] 4 = Likely
- [ ] 5 = Very Likely

6. If you were to consider switching banks, what would be your PRIMARY reason? (Select ONE)

- ○ High or unfair fees
- ○ Better Interest rates elsewhere
- ○ Lack of nearby branches or ATMs
- ○ Poor customer service
- ○ Better mobile / digital tools elsewhere
- ○ I would not consider switching

7. Which of the following would make it difficult for you to switch to a new bank? (Select ALL that apply)

- [ ] Updating direct deposit with my employer
- [ ] transferring automatic bill payments
- [ ] Getting new checks and a new debit card
- [ ] Losing my long-standing bank relationship
- [ ] Not knowing which bank would be better
- [ ] None, I could switch easily

8. How long have you been a customer at your current primary bank?

- [ ] Less than 1 year
- [ ] 1 - 3 years
- [ ] 3 - 7 years
- [ ] 7 - 15 years
- [ ] More than 15 years

9. How many different banks do you currently hold accounts with?

- ○ Only 1 - my primary bank
- ○ 2 banks
- ○ 3 banks
- ○ 4 or more banks

10. Have you switched your primary bank within the last 5 years?

- ○ Yes
- ○ No

1. The reason I chose these survey items is to learn about how customers leave their bank, why they leave and what they want to stay with their bank. The focus is on the four main service areas that have the most significant impact on customers overall experience of being a part of their bank. Therefore, these four service areas are vital for banks to build valid relationships based on trust with customers. The likelihood to recommend question is a general measure of customer loyalty. If a customer would refer someone to their bank, then it is much more likely that they will not switch. There were a number of switching questions included in the surveys. Points. These included primary reason for switching, barriers to switching, length of relationship, number of banks used, and recently switched. The goal of all of these questions was to access the dissatisfaction with and cost of switching. While some customers will not stay with their bank because of dissatisfaction, the cost associated with switching are also present and would therefore create obstacles to switching. These types of costs, such as updating direct deposit information or making changes to automatic payment organization, also present obstacles to switching and help increase retention. By asking customers about their satisfaction, loyalty level, and past behaviors, we are able to analyze the data collected from the database to determine which customers are either currently at risk of leaving or have more significant issues, impacting their ability or willingness to stay with their bank. This database created through the survey is a great value in using predictive, modeling and developing customer retention strategy.

2. Survey items have been developed to have the structure needed for direct integration with relational database, using quantifiable data types. Examples of how to define customer satisfaction include fees, interest rates, digital banking experience, and customer service. Each of these can be rated on a five-point scale. All questions related to satisfaction will have integer values of 1 – 5. As a result, all statistical analyses of averages and trends can be done and correlated to people's likelihood to switch banks. Likelihood to recommend is also a five-point scale and can be treated as an integer variable. This classification will also allow us to define loyalty and allow comparison across segments. The main reason that someone would switch banks will be defined as a single categorical variable and will also be defined using integer values based on a lookup table. The catalogue can also be seen as a list of potential reasons why records are consistent. At the time of asking about barriers to switching, an individual can provide multiple responses to the same barrier, making it difficult for an individual to keep track of barriers when tracking them. For each barrier there could be fields to keep track of barriers independently where 1 means for selected and 0 for not selected or related tables could handle the relationship in a money-to-many correlations between the two records (both the barrier type data and the response to the survey). Each of these solutions has advantages and disadvantages when making a decision on which approach is best where an analysis of several criteria should consider the impact of each alternative will have on seeking additional financial support from sources outside the organization also be completed. Customers will be able to respond to questions about how many banks have they been with previously using original categorical variables, which would allow for ranking and trend analysis. In addition, this will be able to indicate their recent behavior based on switching to a new bank as a binary variable, which is yes or no that can be used to predict potential loss of customer out of future date.

Hello again!

We at Noble Bank (as well as any other bank) use historical data to create profiles of applicants to determine who will get a loan and who will not. However, this is not all we use data for.

We also use data to determine how much they can change product offerings and not lose existing customers. Because there is a certain amount of effort involved with moving your entire checking account to a new bank (new checks, new debit cards, automatic deposits to be updated, and more) banks hold a certain amount of staying power in the inconvenience of "change" (no pun intended).

I would like for you to help me with 2 things:

1. Create a survey and collect some data for me on this issue. Create a survey without at least 10 items to capture data regarding customers decision to stay with or leave their bank. If you have trouble coming up with your items, drop by your local bank and ask the manager what information would be helpful! After you design your survey, invite a few friends, family members, roommates, classmates, anyone you can find to complete your survey.

2. Develop an Entity Relationship Diagram (ERD) of our bank's database to help us re-engineer our database.

As always, I am very much looking forward to your work!

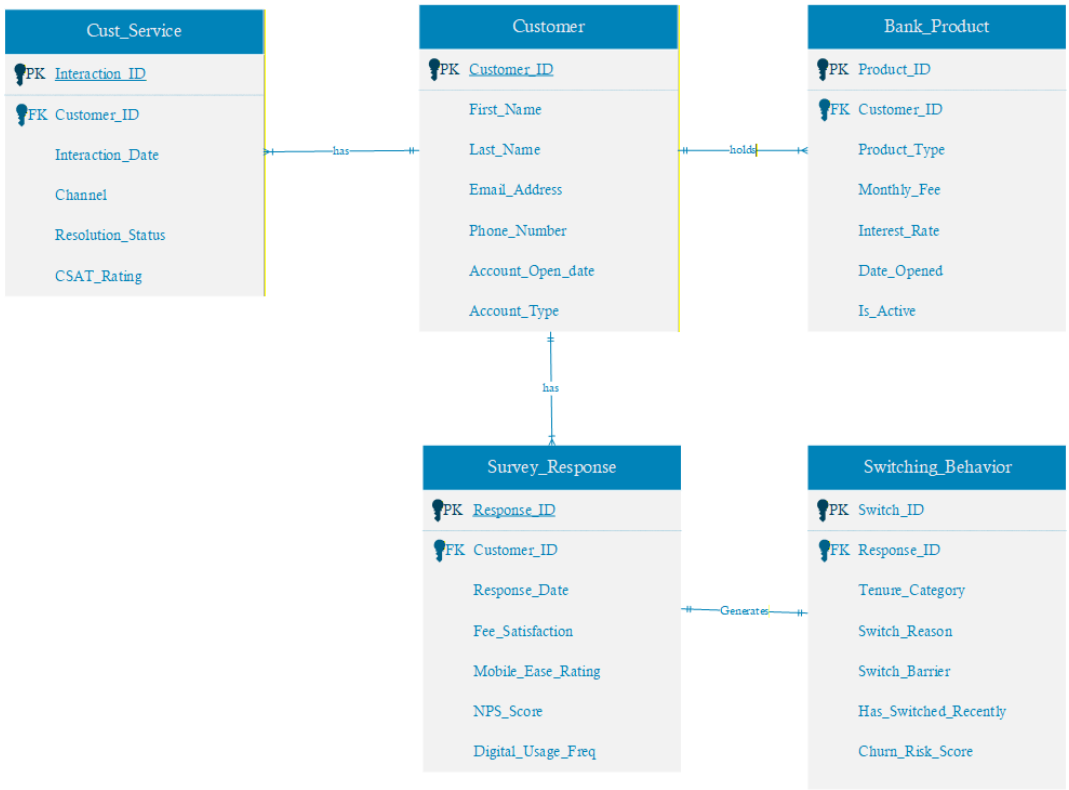
Thanks,
Jeremy

Part 2 Data Management

Select any one functional area in your organization. Think about the data involved in the processes. Based on your selection, use an **Entity Relationship Diagram (ERD)** to depict the data model of the functional area. The examples covered in class may be useful in helping you select a functional area for analysis. Please be sure to select something that is manageable. Please also have 4 entities in your ERD. There is no need to have all possible attributes. <u>But please have relevant/suitable ones for each entity</u>. Some attributes can be pointless, depending on the design of your database. And we don't want to have unnecessary ones.

Your ERD must be constructed using Visio (or equivalent). Paste the ERD in a word document **as a picture** for submission. If you do not have access to Visio, you can conceptualize it and use the computer in the lab. This will minimize the amount of time you need using the software itself.

Submit your work on D2L as a single word document (or pdf) by the due date.



Part 3: Data Queries

ManyBooks.com was developed by JK Rowling as an open marketplace for books online, a one-stop e-commerce search engine where you can search through more than 100 million new, used, rare, and out-of-print books for sale.

Ms. Rowling understands that you have expertise in databases. She needs help developing some custom queries for her sales and marketing team members. She has provided you with a sample data file, DA233_S1X-II.accdb. Specifically, what Jane needs is:

1. Create a table showing the telephone number of each author. Streamline your database so you don't have too much redundancy. Be sure to include the input mask for the telephone number. Make sure the formats are correct. You may make up your own telephone numbers. Save the table, giving it and the fields appropriate names. Hint: The input mask can be set if you choose the text data type for telephone number. (2 pts)
2. Create relationships for the tables and enforce referential integrity. You should be able to generate the necessary relationships for all tables. Hint: To create relationships, click database tools at on the ribbon and you'll see relationships. Drag the tables you want into the design canvas and build the relationships accordingly. You should be able to tell where the relationships are based on the fields. (2 pts)
3. A query of all authors who have written a book that costs more than $10 and is a novel. The query needs to have the author's First Name, Last Name, Title of Book, Price, Category, and Rep displayed. Save the query as **AuthorsQuery**. (2 pts)
4. A query that calculates a 15 percent increase in the purchase price of each book. Sort the query on the new calculated field. Save the query as **BookPricesQuery**. (3 pts)
5. A query that updates the price of each book by subtracting 50 cents. Save the query as **Less50CentsQuery**. (3 pts)

Upload your database (with completed queries) on D2L by the due date.

Rubric for Assignment II

Criteria	Level 5	Level 4	Level 3	Level 2	Level 1
Part 1 and 2: Addressing the requirements and analysis	13 points Exceeds expectations. The solution is well crafted, professional, innovative, and presentable. It shows considerable effort and has potential for further development. The survey is well crafted and neatly laid out.	10 points Meets all criteria. It's presentable and professionally prepared. There is nothing wrong but nothing really outstanding either. A little more research and reflection would be ideal to enhance the solution. It is a decent piece of work nonetheless.	7 points One or more criteria are not met, or some parts are missing. While some parts are ok, others are not.	4 points Several criteria or functions are missing; or several parts are missing.	1 point Completely irrelevant
Part 3 Rubric: Database queries	12 points Perfect solution. Consideration given to usability. Went beyond technical aspects.	9 points Meets most of criteria. There is nothing wrong and it works. But there was little, or no consideration given to the usability/business aspects. The presentation of the solution can be enhanced.	6 points Technically feasible. Did not consider the usability of the solution in some ways. And/Or part of the solution was not technically feasible/correct.	4 points Irrelavant. Or most of the solution was incorrect.	2 points Completely irrelevant